Multi Area OSPF Network

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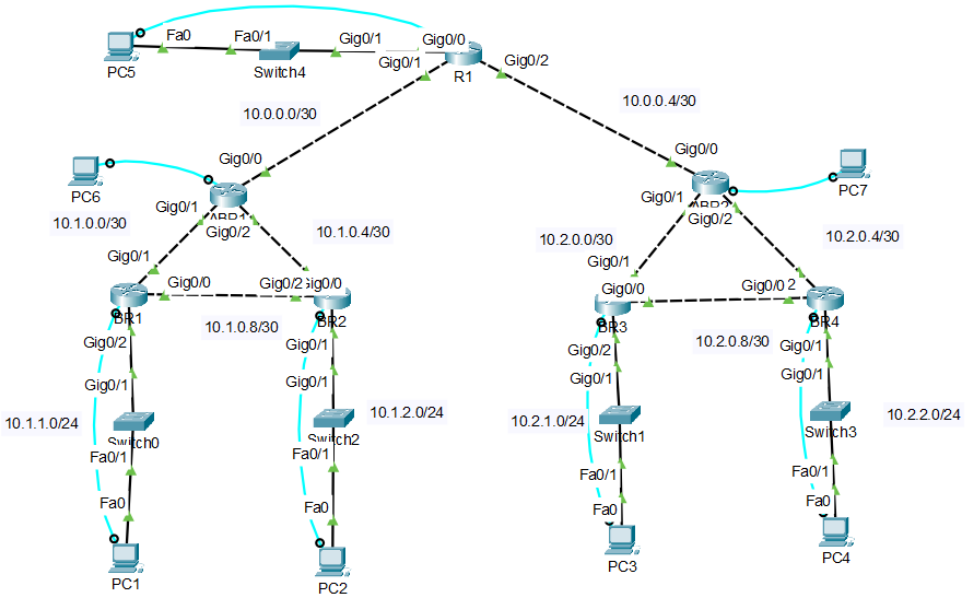
Overview

For this project I built a multi area OSPF network in Cisco Packet Tracer for a company with a head office and two branches. My goal was to understand how OSPF areas work, how Area Border Routers (ABRs) connect each branch to the backbone, and how route summarization keeps routing tables small.

In the finished network every PC can reach every other PC and the head office router, and each branch is advertised into the backbone as one summary route. I checked everything with pings and traceroutes between the areas.

Topology

The network has three OSPF areas. Area 0 is the backbone with R1 in the middle. Area 1 (Branch A) hangs off ABR1 and Area 2 (Branch B) hangs off ABR2. I planned the addresses so that everything in Branch A starts with 10.1 and everything in Branch B starts with 10.2, which is what makes the summarization work.



· **Area 0 (Backbone):** R1 connects to ABR1 (10.0.0.0/30) and ABR2 (10.0.0.4/30). R1 also has the HQ LAN (10.0.10.0/24) behind Switch4 and a loopback of 1.1.1.1.

· **Area 1 (Branch A):** ABR1 connects to BR1 (10.1.0.0/30) and BR2 (10.1.0.4/30). BR1 and BR2 are also linked to each other (10.1.0.8/30). PC1 is on 10.1.1.0/24 and PC2 is on 10.1.2.0/24.

· **Area 2 (Branch B):** Same layout with 10.2 addresses. ABR2 connects to BR3 and BR4, BR3 and BR4 are linked, and PC3 and PC4 are on 10.2.1.0/24 and 10.2.2.0/24.

· **Management:** PC5 sits on the HQ LAN and also has a console cable to R1. PC6 and PC7 are console machines for ABR1 and ABR2.

· **Router IDs:** every router has a loopback that OSPF uses as its router ID, from 1.1.1.1 on R1 up to 7.7.7.7 on BR4.

IP Addressing

Device	Interface	IP Address	Connects To
R1	Lo0	1.1.1.1/32	Router ID
R1	G0/0	10.0.10.1/24	HQ LAN (Switch4)
R1	G0/1	10.0.0.1/30	ABR1
R1	G0/2	10.0.0.5/30	ABR2
ABR1	Lo0	2.2.2.2/32	Router ID
ABR1	G0/0	10.0.0.2/30	R1
ABR1	G0/1	10.1.0.1/30	BR1
ABR1	G0/2	10.1.0.5/30	BR2
ABR2	Lo0	3.3.3.3/32	Router ID
ABR2	G0/0	10.0.0.6/30	R1
ABR2	G0/1	10.2.0.1/30	BR3
ABR2	G0/2	10.2.0.5/30	BR4
BR1	Lo0	4.4.4.4/32	Router ID
BR1	G0/0	10.1.0.9/30	BR2
BR1	G0/1	10.1.0.2/30	ABR1
BR1	G0/2	10.1.1.1/24	PC1 LAN (Switch0)
BR2	Lo0	5.5.5.5/32	Router ID
BR2	G0/0	10.1.0.10/30	BR1
BR2	G0/1	10.1.2.1/24	PC2 LAN (Switch2)
BR2	G0/2	10.1.0.6/30	ABR1
BR3	Lo0	6.6.6.6/32	Router ID
BR3	G0/0	10.2.0.9/30	BR4
BR3	G0/1	10.2.0.2/30	ABR2
BR3	G0/2	10.2.1.1/24	PC3 LAN (Switch1)
BR4	Lo0	7.7.7.7/32	Router ID
BR4	G0/0	10.2.0.10/30	BR3
BR4	G0/1	10.2.2.1/24	PC4 LAN (Switch3)
BR4	G0/2	10.2.0.6/30	ABR2

Each PC uses .2 in its LAN with the router's .1 as its default gateway, for example PC1 is 10.1.1.2 with a gateway of 10.1.1.1.

How the design works

Each branch is its own OSPF area, and the only way between areas is through Area 0. ABR1 and ABR2 each have one interface in Area 0 and the rest in their branch area, so they are the ones that pass routes between the branch and the backbone.

Inside each branch, the two branch routers are linked to each other. This means traffic between PC1 and PC2 goes straight from BR2 to BR1 and never has to touch the ABR, and it gives each branch router a second way to reach its ABR.

I set `auto-cost reference-bandwidth 100000` on every router. By default OSPF treats anything 100 Mbps or faster as cost 1, so all the gigabit links would look the same. This value has to be the same on every router or the path costs stop making sense. I

also made the loopbacks passive since no OSPF neighbors live on them.

For summarization, ABR1 has `area 1 range 10.1.0.0 255.255.0.0` and ABR2 has `area 2 range 10.2.0.0 255.255.0.0`. Instead of sending five separate routes per branch into the backbone (three /30 links and two LANs), each ABR sends a single /16. The branch loopbacks (4.4.4.4 to 7.7.7.7) are outside these ranges, so they are still advertised on their own.

Issues I ran into

HQ LAN was not in OSPF

At first R1 only advertised its two backbone links and its loopback. I forgot the HQ LAN, so the branches had no route to 10.0.10.0/24. Adding `network 10.0.10.0 0.0.0.255 area 0` on R1 fixed it.

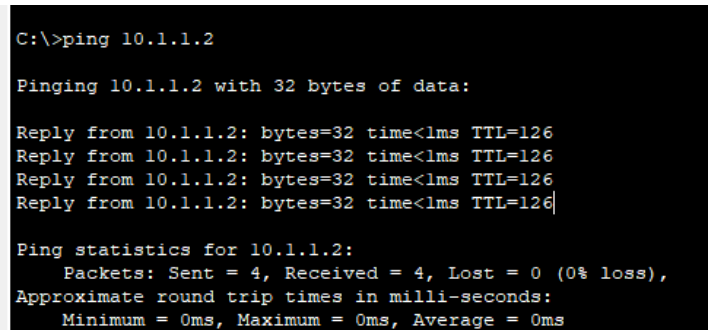
Final state

Every router advertises its own networks and loopback, the HQ LAN is reachable from both branches, and both ABRs summarize their branch into a single /16 route toward the backbone.

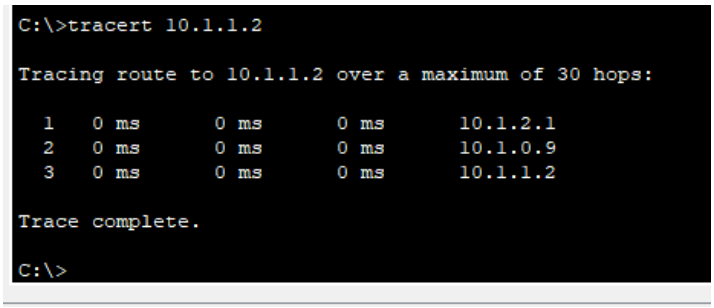
Results

PC2 to PC1 (inside Area 1)

The traceroute goes 10.1.2.1, then 10.1.0.9, then PC1. Traffic uses the direct BR2 to BR1 link and stays inside Area 1.



Ping



Traceroute

PC4 to PC3 (inside Area 2)

Same result in Branch B: 10.2.2.1, then 10.2.0.9, then PC3. The first ping timed out because PC4 was still using ARP to find the MAC address, which is normal for the first packet.

```
C:\>ping 10.2.1.2

Pinging 10.2.1.2 with 32 bytes of data:

Request timed out.
Reply from 10.2.1.2: bytes=32 time<1ms TTL=126
Reply from 10.2.1.2: bytes=32 time<1ms TTL=126
Reply from 10.2.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 10.2.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping

```
C:\>tracert 10.2.1.2

Tracing route to 10.2.1.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.2.2.1
  2    0 ms      0 ms      0 ms      10.2.0.9
  3    0 ms      0 ms      0 ms      10.2.1.2

Trace complete.
```

Traceroute

PC1 to R1 (Area 1 to the backbone)

PC1 reaches R1's loopback 1.1.1.1 through BR1 and ABR1: 10.1.1.1, then 10.1.0.1, then 1.1.1.1.

```
C:\>ping 1.1.1.1

Pinging 1.1.1.1 with 32 bytes of data:

Reply from 1.1.1.1: bytes=32 time<1ms TTL=253
Reply from 1.1.1.1: bytes=32 time<1ms TTL=253
Reply from 1.1.1.1: bytes=32 time<1ms TTL=253
Reply from 1.1.1.1: bytes=32 time<1ms TTL=253

Ping statistics for 1.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping

```
C:\>tracert 1.1.1.1

Tracing route to 1.1.1.1 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.1.1.1
  2    0 ms      0 ms      0 ms      10.1.0.1
  3    0 ms      0 ms      0 ms      1.1.1.1

Trace complete.
```

Traceroute

PC4 to R1 (Area 2 to the backbone)

PC4 reaches 1.1.1.1 through BR4 and ABR2: 10.2.2.1, then 10.2.0.5, then 1.1.1.1.

```
C:\>ping 1.1.1.1

Pinging 1.1.1.1 with 32 bytes of data:

Reply from 1.1.1.1: bytes=32 time<1ms TTL=253
Reply from 1.1.1.1: bytes=32 time<1ms TTL=253
Reply from 1.1.1.1: bytes=32 time<1ms TTL=253
Reply from 1.1.1.1: bytes=32 time<1ms TTL=253

Ping statistics for 1.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping

```
C:\>Tracert 1.1.1.1

Tracing route to 1.1.1.1 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.2.2.1
  2    0 ms      0 ms      0 ms      10.2.0.5
  3    0 ms      0 ms      0 ms      1.1.1.1

Trace complete.
```

Traceroute

PC4 to PC1 (Branch B to Branch A)

This is the longest path in the network: 10.2.2.1, 10.2.0.5, 10.0.0.5, 10.0.0.2, 10.1.0.2, then PC1. It goes from Area 2 into the backbone, through R1, and back out into Area 1, which shows that traffic between areas always passes through Area 0.

```
C:\>ping 10.1.1.2

Pinging 10.1.1.2 with 32 bytes of data:

Reply from 10.1.1.2: bytes=32 time<1ms TTL=123
Reply from 10.1.1.2: bytes=32 time<1ms TTL=123
Reply from 10.1.1.2: bytes=32 time<1ms TTL=123
Reply from 10.1.1.2: bytes=32 time<1ms TTL=123

Ping statistics for 10.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Ping

```
C:\>tracert 10.1.1.2

Tracing route to 10.1.1.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      10.2.2.1
  2    0 ms      0 ms      0 ms      10.2.0.5
  3    0 ms      0 ms      0 ms      10.0.0.5
  4    0 ms      0 ms      0 ms      10.0.0.2
  5    0 ms      0 ms      0 ms      10.1.0.2
  6    0 ms      0 ms      0 ms      10.1.1.2

Trace complete.
```

Traceroute

What I learned

· OSPF areas keep things organized. The branch routers only need the details of their own area, and the ABRs handle everything else.

· Traffic between two branch areas always goes through Area 0. The PC4 to PC1 traceroute shows this clearly.

· Planning the IP addresses per area (10.1 for Branch A, 10.2 for Branch B) is what made summarization a single command on each ABR.

· The reference bandwidth has to be the same on every router.

Appendix: Final Configurations

R1

```
en
conf t
ho R1
no ip domain-lookup
line vty 0 15
logging sync
exit
int l0
ip add 1.1.1.1 255.255.255.255
no shut
int g0/1
ip add 10.0.0.1 255.255.255.252
no shut
int g0/2
ip add 10.0.0.5 255.255.255.252
no shut
int g0/0
ip add 10.0.10.1 255.255.255.0
no shut
ex
router ospf 1
network 10.0.0.0 0.0.0.3 area 0
network 10.0.0.4 0.0.0.3 area 0
network 1.1.1.1 0.0.0.0 area 0
network 10.0.10.0 0.0.0.255 area 0
auto-cost reference-bandwidth 100000
passive-int l0
ex
do copy run start
```

ABR1

```
en
conf t
ho ABR1
no ip domain-lookup
line vty 0 15
logging sync
exit
int l0
ip add 2.2.2.2 255.255.255.255
no shut
int g0/1
ip add 10.1.0.1 255.255.255.252
no shut
```

```
int g0/2
ip add 10.1.0.5 255.255.255.252
no shut
int g0/0
ip add 10.0.0.2 255.255.255.252
no shut
ex
router ospf 1
network 10.1.0.0 0.0.0.3 area 1
network 10.1.0.4 0.0.0.3 area 1
network 2.2.2.2 0.0.0.0 area 0
network 10.0.0.0 0.0.0.3 area 0
auto-cost reference-bandwidth 100000
passive-int l0
area 1 range 10.1.0.0 255.255.0.0
ex
do copy run start
```

ABR2

```
en
conf t
ho ABR2
no ip domain-lookup
line vty 0 15
logging sync
exit
int g0/1
ip add 10.2.0.1 255.255.255.252
no shut
int g0/2
ip add 10.2.0.5 255.255.255.252
no shut
int g0/0
ip add 10.0.0.6 255.255.255.252
no shut
int l0
ip add 3.3.3.3 255.255.255.255
no shut
ex
router ospf 7
network 10.2.0.0 0.0.0.3 area 2
network 10.2.0.4 0.0.0.3 area 2
network 3.3.3.3 0.0.0.0 area 0
network 10.0.0.4 0.0.0.3 area 0
auto-cost reference-bandwidth 100000
passive-int l0
area 2 range 10.2.0.0 255.255.0.0
ex
do copy run start
```

BR1

```
en
conf t
ho BR1
no ip domain-lookup
line vty 0 15
logging sync
exit
int l0
ip add 4.4.4.4 255.255.255.255
no shut
int g0/2
ip add 10.1.1.1 255.255.255.0
no shut
ex
int g0/0
ip add 10.1.0.9 255.255.255.252
no shut
int g0/1
ip add 10.1.0.2 255.255.255.252
no shut
ex
router ospf 4
network 10.1.1.0 0.0.0.255 area 1
network 10.1.0.8 0.0.0.3 area 1
network 4.4.4.4 0.0.0.0 area 1
network 10.1.0.0 0.0.0.3 area 1
auto-cost reference-bandwidth 100000
passive-int l0
ex
do copy run start
```

BR2

```
en
conf t
ho BR2
no ip domain-lookup
line vty 0 15
logging sync
exit
int l0
ip add 5.5.5.5 255.255.255.255
no shut
int g0/1
ip add 10.1.2.1 255.255.255.0
no shut
int g0/0
ip add 10.1.0.10 255.255.255.252
no shut
int g0/2
ip add 10.1.0.6 255.255.255.252
no shut
ex
router ospf 5
```

```
network 10.1.2.0 0.0.0.255 area 1
network 10.1.0.8 0.0.0.3 area 1
network 5.5.5.5 0.0.0.0 area 1
network 10.1.0.4 0.0.0.3 area 1
auto-cost reference-bandwidth 100000
passive-int l0
do copy run start
```

BR3

```
en
conf t
ho BR3
no ip domain-lookup
line vty 0 15
logging sync
exit
int l0
ip add 6.6.6.6 255.255.255.255
no shut
int g0/2
ip add 10.2.1.1 255.255.255.0
no shut
int g0/0
ip add 10.2.0.9 255.255.255.252
no shut
int g0/1
ip add 10.2.0.2 255.255.255.252
no shut
ex
router ospf 6
network 10.2.1.0 0.0.0.255 area 2
network 10.2.0.8 0.0.0.3 area 2
network 6.6.6.6 0.0.0.0 area 2
network 10.2.0.0 0.0.0.3 area 2
auto-cost reference-bandwidth 100000
passive-int l0
ex
do copy run start
```

BR4

```
en
conf t
ho BR4
no ip domain-lookup
line vty 0 15
logging sync
exit
int l0
ip add 7.7.7.7 255.255.255.255
no shut
int g0/1
ip add 10.2.2.1 255.255.255.0
no shut
int g0/0
```

```
ip add 10.2.0.10 255.255.255.252
no shut
int g0/2
ip add 10.2.0.6 255.255.255.252
no shut
ex
router ospf 7
network 10.2.2.0 0.0.0.255 area 2
network 10.2.0.8 0.0.0.3 area 2
network 10.2.0.4 0.0.0.3 area 2
network 7.7.7.7 0.0.0.0 area 2
auto-cost reference-bandwidth 100000
passive-int l0
ex
do copy run start
```